

Numerical study of microalgae cultivation process in a batch liquid foam-bed photobioreactor using CFD approach

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ARTICLE INFO

Keywords:

Computational fluid dynamic simulation
Microalgae cultivation
Liquid foam-bed photobioreactor
Fluid phase behavior
Light effect

ABSTRACT

Microalgae have different uses as feedstock of beneficial chemicals such as polysaccharides, carotenoids, and polyunsaturated fatty acids. Furthermore, they have been applied in the renewable energy, food, agriculture and feed industries, and in wastewater treatment. The growth and cultivation of microalgae are highly regarded by researchers worldwide. So, a variety of systems and designs have been used to cultivate microalgae. A research team recently proposed a new experimental design of photobioreactors based on a liquid foam flow at Wageningen University in the Netherlands. In this study, a numerical study of hydrodynamics and mass transfer of this novel system was conducted using computational fluid dynamics (CFD). This method is based on the simultaneous solution of transfer equations including hydrodynamic equations, mass transfer, cell growth, and radiation transport equation. The simulation was performed in an unsteady and three-dimensional state in COMSOL 5.5 software. The flow regime was considered laminar and mixture model and transport dilution species physics were used to model the behavior of two-phase fluid. Manual geometry grid generation was performed after defining the velocity and concentration boundaries. The number of optimal meshes was determined to be approximately 1.2×10^5 using the mesh independence test. The simulation results were validated according to the concentration-time curve. The cell density, 22 g l^{-1} , is minimal in the center. The maximum biomass is 30.9 g l^{-1} at the bed wall after 80 h at $1446 \mu\text{mol photon. m}^{-2} \text{ s}^{-1}$ light flux for the minimal cell density.

1. Introduction

Liquid foam beds are applied for the removal of pollutants through physico-chemical methods despite not being used in large scale [1]. These reactor types are beneficial for the absorption of contaminant in the gas phase due to their large interfacial area in which high mass transfer rates can be reached. This facilitates the pollutant treatment from the gas phase to the liquid phase. Undesirable materials are absorbed to the liquid flow and can be converted through chemical reactions. Aside from large interfacial area, foam beds are useful since they offer a low pressure drop and enhanced contact time between the liquid and gas phases [2–4]. The problem of pollutant removal and CO_2 fixation is an important challenge [5]. These reactors can also be used in bioprocess operations [6]. Similarly, wet foam bioreactors are also in use in order to remove the

pollutants. In these bioreactors, instead of chemical interactions, the dispersed microorganisms in the foam flow and bio surfactants are important for the biodegradation [7,8]. The biodegrading microorganisms are quickly growing at cell culture in the liquid phase of the foams. Additionally, for foam bed bioreactors, the benefit relies on the large surface area of the foam leading to high contaminant treatment rates. Organic pollutants can be absorbed using an organic-phase dispersed in the liquid foam flow [5,9]. With the goal of achieving high cell densities, advanced and novel photobioreactor technologies are proposed. These comprise biofilm photobioreactors [10], thin flat-panels photobioreactor [11], thin-layer photobioreactors [12], and photobioreactors based on falling droplets [13]. An analogy can be drawn among these photobioreactors in terms of algal cell density, the ability of the mass transfer, energy requirements, and operational costs.

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<https://doi.org/10.1016/j.renene.2024.120054>

Received 9 September 2022; Received in revised form 5 January 2024; Accepted 23 January 2024

Available online 30 January 2024

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The enhancement of simulation tools and methods can improve the design of the photobioreactor and help the successful scale up procedure of the photobioreactor in addition to providing numerical predictions for its performance. The development of such a comprehensive model requires the definition of appropriate sub-models and the provision of the correct method for coupling these sub-models [14]. A comprehensive simulation approach to photobioreactors involves at least three sub-models: (1) a descriptive hydrodynamic model for mixing liquid (culture medium), solid (biomass), and gas (O_2 and CO_2 bubbles) phases, (2) a radiation transport model, and (3) a model for microalgae biological growth kinetics. Sub-models associated with photosynthesis typically emphasize the limitation imposed by light and several environmental factors such as the effects of temperature, change in the pH, and improving the growth medium composition [15].

Many researchers have proposed different approaches to develop multiphase models. In a study by Gao et al., three different phases were suggested for effective simulation of fluid motion [16]. Computational approaches for simulating gas bubbles dispersed in a liquid include methods such as the volume of fluid (VOF), Eulerian-Lagrangian, and Eulerian-Eulerian. In the Eulerian-Lagrangian approach, the liquid phase is the continuous phase, in which a force balance is applied to analyze the gas flow behavior and determine the bubble mobility pathways [16–18]. The behavior of the fluid field in the photobioreactors has been evaluated by several researchers such as Zhao et al. [19], Patil et al. [20], Cho et al. [21], Belohlav et al. [22], and Rosales et al. [23].

Likewise, radiation transport models can vary in the design of photobioreactor processes. The simplest and most widely used equation for radiation transport is represented by the Beer-Lambert law [16]. A more accurate method for solving the general radiative transfer equation in three dimensions was introduced by Kong and Vigil, which requires precise information about the culture medium for model definition [17]. In photobioreactors, the flux of photons decreases in proportion to the distance to the light source. This phenomenon of reduced light penetration becomes more prominent when the photobioreactors are tested at high concentrations of cell growth. In the previous studies, three prevalent models have been discussed for modeling the light transmission to the bioreactor, including the Beer-Lambert law, two-flux approximation, and the general radiative transfer equation [24].

An experimental system design was proposed to achieve goals such as increasing the biomass production of *Chlorella sorokiniana* and reducing the energy consumption of process equipment [24]. This photobioreactor utilizes a two-phase liquid foam flow to enhance contact between the liquid and gas phases and improve the mass transfer between them [24,25]. Despite the growing interest in algae cultivation, mass production levels have not yet been reached. Therefore, this study aims to utilize Computational Fluid Dynamics (CFD) to simulate mass transport phenomena and hydrodynamics in a batch Liquid Foam Bed Photobioreactor (LFBPBR) system with the presence of *Chlorella Sorokiniana* microalgae strain. No computational analysis has been conducted using the CFD method for designing the intended photobioreactor or similar designs. Additionally, there have been limited number of studies on the few simultaneous coupling of mass transfer equations and hydrodynamics, cell growth kinetic and radiation transport in the reactors containing liquid foam flows; therefore, microalgae production feasibility in a novel photobioreactor using computer simulation is one of the considerable aspects of this study. A batch LFBPBR has many privileges compared to common liquid-phase photobioreactors for microalgae cultures. These privileges include:

- (1) Enhancement of gas transfer rate
- (2) Reduced gassing energy requirement
- (3) Low hydrostatic pressure
- (4) Short radiation path and high cell densities production [25–27].

The unit of concentration of produced biomass is g/l for the specific photobioreactor with constant reactor volume or cultivation system. It is clear that in the scale up process of photobioreactor evaluating dimensionless number is crucial. For example, considering ratio of concentration of produced biofuel to concentration of cultivation system ($(\text{concentration of produced biofuel (g/l)})/(\text{concentration of cultivation system (g/l)})$) can help to evaluate and compare photo bioreactor in the scale up or large size with other types of reactors. However, It is proved that concentration biofuel produced in the foam-bed photobioreactor is noticeably higher than liquid phase photobioreactor. Considering the mentioned dimensionless number, the concentration of biofuels will increase with the concentration of biomass in the culture medium. However, for comparison with conventional cultivation systems it is more important to compare the amount of biomass produced in relation to the size of the cultivation system, since in the case of the Foam-bed reactor high concentrations of biomass in the cultivation medium can be achieved, but in principle there is a small amount of high volume cultivation medium in the cultivation system. Foam-bed photobioreactors significantly decrease the energy needed for gas supply. This reduction occurs because the foam-bed's hydrostatic pressure is notably lower compared to an equivalent water column, reducing the necessity for intense gas compression. The foam-bed's reduced pressure also decreases the gas flow, necessitating lower superficial gas velocities than those in liquid photobioreactors. This is due to the foam-bed's higher gas transfer capacity resulting from increased interfacial area. Hence, the hydrostatic pressure reduction in the photobioreactor affects the energy required for gas supply in the reaction medium. Moreover, the enhanced mass transfer in the foam-bed photobioreactor is due to increased surface area between gas and liquid phases and prolonged contact time between CO_2 -enriched gas and microalgae-containing liquid. Additionally, the reduced energy needed for gas supply is a result of decreased pressure drop due to the foam's lower density compared to liquid. In essence, the reduced hydrostatic pressure in foam-bed photobioreactors, in contrast to liquid bed photobioreactors, is due to the foam's lower density instead of a liquid phase. In larger-scale photobioreactors, utilizing a recycling pump to make the medium more conducive for transport phenomena, especially the mass transfer coefficient, is a common practice. Introducing a recycling path in the foam-bed photobioreactor could provide a potential solution.

The photobioreactors simulation literature review shows a lack of any numerical or computational analyses regarding the hydrodynamic and mass transport simulation of a LFBPBR using computational fluid dynamics, while CFD can provide more detailed information regarding the behavior of two-phase fluids in these bioreactors. This study employs the coupling and simultaneous solving of four equation groups for the fluid flow including, mass transfer, microalgae cell growth kinetics, and light transfer sub-models to predict the performance of an advanced photobioreactor using CFD technique.

2. Materials and methods

2.1. Experimental review

The system studied in this work is a two-phase photobioreactor based on a novel experimental design proposed by Janoska et al. [25] which operates batch at a constant temperature of $37^\circ C$ and in an unsteady state. The inlet flow to the reactor is a mixture of O_2 , CO_2 , and N_2 in an aqueous solvent (as the predominant component of the culture medium). Thus, the two-phase mixture consists of gas (or air) phase and water phase. The photobioreactor is a vertical glass cylinder 10.4 cm in diameter and 50 cm height. Fig. 1 shows a schematic of the LFBPBR.

Table 1 presents the descriptions of the different sections in Fig. 1a. Since cultured microalgae are autotrophs, organic carbon sources are not required to feed the microorganisms, and CO_2 is the main carbon

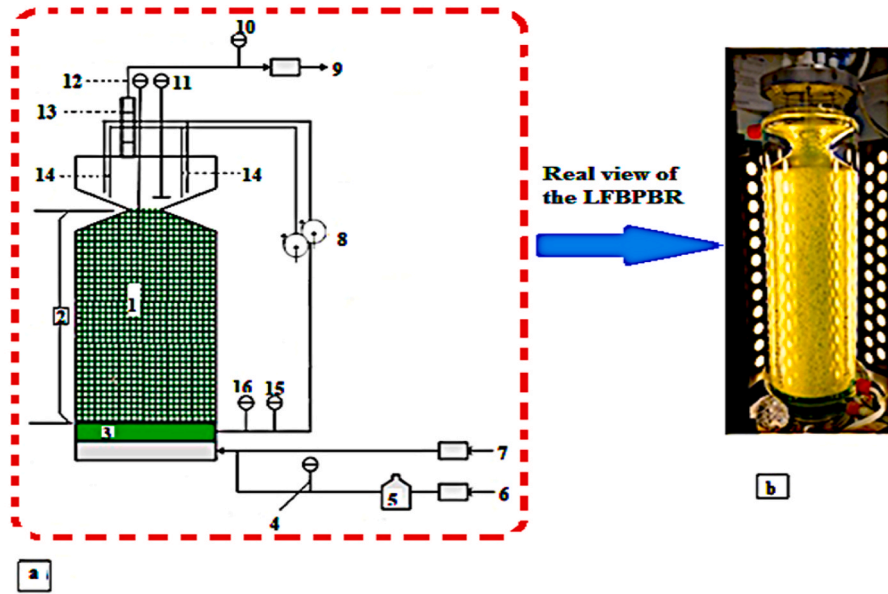


Fig. 1. Two-dimensional (a) and real (b) views of the liquid foam-bed photobioreactor [25].

source for cells growth. In the inlet air, O_2 and CO_2 enter with a certain volume percent and 1 g l^{-1} of the initial microalgae is present as the starting material for cell growth in the bed. The most important physical parameters defined in the LFBPBR are given in Table 2.

Considering the nature of photosynthesis in autotroph strains such as *Chlorella sorokiniana*, the energy required for growth is received through light energy. So, LED lamps, as an artificial light source, are placed around the bed in the system. Photosynthesis initiates when the gas feed is injection into the bed and the light energy is irradiated. Photosynthesis progresses over time and causes the formation of biomass of the *Chlorella sorokiniana* strain along with by-products such as O_2 and carbohydrates. High-pressure injection of gas flow into the bed leads to the formation of spherical bubbles. The addition of some surfactant to the inoculum microalgae leads to a massive flow of spherical foams is formed, creating a relatively greater tendency for them to form in the bed due to the reduced surface tension of the fluid [27]. A bypass flow is implemented to recycle the culture medium flow from the bottom of the bioreactor to the top, and this recycling flow is injected into the bed through its inlets. Bioconversion occurs, the species involved in this biochemical interaction, i.e., microalgal cell mass, CO_2 , and O_2 are consumed or produced between two phases, and their concentration changes by the contact of gas and liquid phases (usually adjacent to two or more foams component). Therefore, in this study, it is crucial to predict the fluid flow field and component concentrations changes.

2.2. Mathematic modeling

2.2.1. Coupling methodology

The problem is modeled in two stages. The fluid flow is initially described by differential equations in terms of the velocity and pressure field. The boundary conditions in the solution range are then established. In the second stage, the mass transfer equation is defined; this is done via the mechanisms of convection and diffusion, as well as the transfer of the major species between two phases. It should be noted that the desired equation is solved by incorporating the velocity field obtained in the first stage. Next, the Beer-Lambert equation is used to determine the distribution of light absorbed by the microalgal cells. Therefore, it can be stated that one-way coupling has been employed twice: once for solving the concentration-velocity coupling equation and once for solving the concentration-light equation coupling.

2.2.2. Assumptions

In this study, a liquid foam-bed photobioreactor proposed by Janoska et al. [25] was simulated using COMSOL Multiphysics software and the finite element method (FEM), the results were compared with experimental results, and the model was validated. The simplifying assumptions were as follows [28]:

1. The predominant compound in the liquid phase is water. But as mentioned by S. Belachqer-El Attar et al. utilization of pure water properties is not correct. At 37°C , viscosity of cultivation system is $1.6755 \text{ [mN s m}^{-2}]$ or $1.6755 \times 10^{-3} \text{ Pa s}$ and the correlation reported S. Belachqer-El Attar et al. were used for the calculation of hydrodynamic of photobioreactor.
2. Considering the Reynolds number, the flow regime is assumed to be laminar.
3. The temperature throughout the process is constant at 37°C , and the absolute pressure of the bed is assumed to be 1 atm;
4. All gases assumed as ideal gases;
5. The dissolution rate of gases in the interface of phases is calculated using Henry's law;
6. Given the small volume fraction of the dispersed phase (liquid phase) compared to the continuous phase, the Euler model are used to simulate both phases [29];
7. The relative velocity vector of two phases has only one component in the direction of height.
8. The direction of radiation transport from the light source to the bioreactor is assumed to be in the direction of the bed radius, and the axial changes of the light are negligible;
9. The Perry model is used to mathematically describe the mass transfer, the Jassby and Platt model [28] is used to describe the kinetics of microalgae growth, and the Beer-Lambert model is used to simulate radiation transport [30].

2.2.3. Hydrodynamic modeling

One of the hydrodynamic models that is appropriate for modeling the multiphase behavior of a fluid in photobioreactors is a mixture model. In this model, a Eulerian model is employed so as to model both continuous and dispersed phases. Eq. (1) reflects the momentum equation that was solved for this purpose:

$$\rho(j \cdot \nabla)j + (\rho_d - \rho_c)(j_{slip} \cdot \nabla)j = \nabla \cdot [-pI + K] - \nabla \cdot K_m + F_m + \rho g + F \quad (1)$$

Table 1

The descriptions of the LFBPBR various sections [25].

Part No.	Description	Part No.	Description
1	The place where the liquid foam forms	9	Gas outlet
2	Computational domain	10	Dew point sensor
3	The initial volume of microalgae	11	Liquid level sensor
4	Dew point sensor	12	Temperature sensor
5	The humidifier of the inlet gas	13	Condenser
6	Inlet N ₂	14	Liquid inlets
7	Inlet O ₂ and CO ₂	15	Location of sampling
8	Special pumps for recycling the liquid phase	16	pH meter

Table 2

Physical characteristics of the liquid foam-bed photobioreactor [25].

Parameter	Value
Bed temperature	37 °C
Inlet gas velocity	0.61 mm s ⁻¹
Inlet fluid velocity	0.59 mm s ⁻¹
Foam radius	0.74 mm
Initial volume fraction of the liquid phase in the foam	0.11
Liquid phase viscosity	1.6755 × 10 ⁻³ Pa s
Liquid phase density	993.21 kg m ⁻³
Bed radius	5.2 cm
Percentage of inlet CO ₂	0.161
Percentage of inlet O ₂	0.1361
Initial algal concentration	1 kg m ⁻³

Table 3

The calculation of the concentration values of gas constituents in the inflow gas.

	X _i : The amount of gas constituent in the inflow gas	MD: Air molar density (mol. m ⁻³)	C _i : The inlet concentration of gas constituent
O ₂	0.161	39.29	6.32
CO ₂	0.1361	39.29	5.34

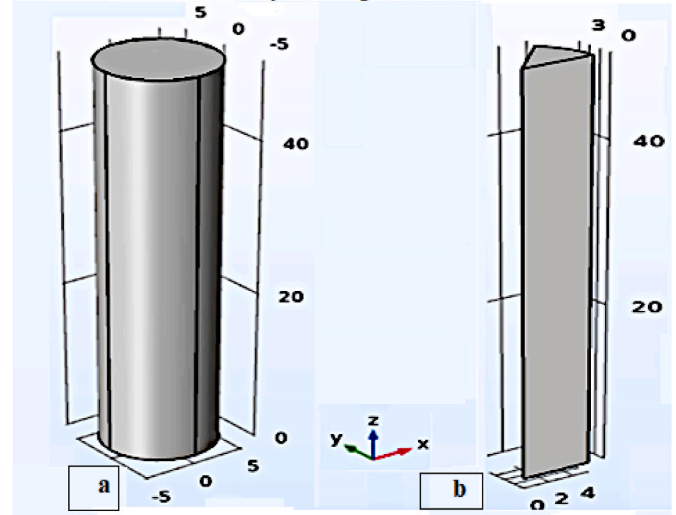
$$\rho = \rho_d \varphi_d + \rho_c (1 - \varphi_d) \quad (2)$$

The continuity equation is obtained through Eq. (3) which m_{dc} indicates the transfer rate of two phases and is assumed equal to zero.

$$\nabla \cdot j = m_{dc} \left(\frac{1}{\rho_c} - \frac{1}{\rho_d} \right) \quad (3)$$

The gas flow enters the bed from the bottom section of the bed (the input boundary), and it leaves

Where j is the quantitative vector velocity of the mixture, p denotes the hydrodynamic pressure, j_{slip} denotes the relative velocity between the two phases of gas and liquid, and g denotes the gravitational acceleration vector. Eq. (2) shows the density of a two-phase mixture, where φ_d is the volume fraction of the dispersed phase, ρ_c is the density of the continuous phase, ρ_d is the density of the dispersed phase, and φ_c is the volume fraction of the continuous phase. K_m and K , which come from Eq. (4), coupled with force F , represents the external force that is acting on the fluid volume. F_m denotes the momentum transfer that is occurring between the two phases. The continuity equation is obtained through Eq. (3). Moreover, there is a transfer equation in the mixture model whose differential form is given in Eq. (5) in which m_{dc} indicates the transfer rate of two phases and is equal to zero in this research because two phases of water (liquid phase) and gaseous phase are non-transferable to each other. φ_d parameter is the volume fraction of the liquid phase.

Geometry of the photobioreactor**Fig. 2.** The 3D schematic of the LFBPBR geometry (a) and 45-degree segment (b).

$$K = \mu (\nabla j + (\nabla j)^T) - \frac{2}{3} \mu (\nabla \cdot j) I, K m = (\rho_c + \rho_d - \rho) u_{slip} j_{slip}^T \quad (4)$$

$$j \cdot \nabla \varphi_d + \nabla \cdot j_{slip} = \frac{m_{dc} \rho}{\rho_d \rho_c} \quad (5)$$

For the fluid velocity component is assumed to be zero in directions other than z direction, the value of the real velocity of the fluid, at the inlet of the photobioreactor is equal to the difference in apparent velocity of the two phases at the inlet ($u_0 = u_{c,in} - u_{d,in}$). There is a no slip condition close to the wall ($u = 0$), and because the bed is thought to be containing just the liquid phase at the beginning of the process, the initial value of the liquid volume fraction will be one ($\varphi_{d,0} = 1$). The fluid of the bed, on the other hand, is discharged at ambient pressure (as a constant value in the boundary condition's outlet). Velocity of the dispersed phase is equal to the intensity of the flow, which is specified as the liquid inlet velocity (see Table 2). However, the initial values inserted to COMSOL in the hydrodynamic model are the same as 1 atm of absolute pressure, and 0.1 vol fraction of liquid phase, according to the information in the reference article [25]. The flow regime in this research was laminar because of the small scale of the foam dimensions (in mm) and the dispersed phase velocities (in mm.s⁻¹).

2.2.4. Mass transfer and cell growth modeling

Within the steady-state and continuous flow inside the LFBPBR, the transfer of chemical components is conducted through diffusion and convection. The physics of transport diluted species (TDS) is employed twice; once to transfer gas constituents from the gas phase to the liquid one and once to contrary exchange. The general equation of mass transfer would be Eq. (6), in the case that the reaction is assumed to take

Table 4

The statistical analysis of the geometry meshing.

Mesh statistics			
Maximum element size	0.523	Prism elements	12,378
Minimum element size (cm ²)	0.157	Tetrahedral elements (no)	109,059
Maximum element growth rate	1.2	Pyramid elements (no)	14,338
Curvature factor (cm ⁻¹)	0.7	Volume of elements (cm ³)	530.7
Resolution of narrow regions (pixel/inch)	0.6	Total elements (no)	121,435
Average quality mesh	0.66		

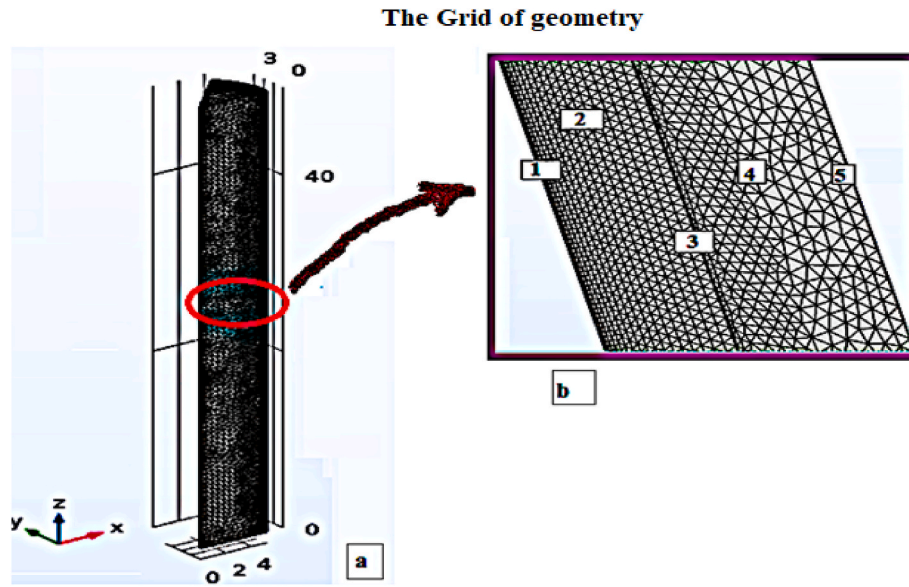


Fig. 3. Three-dimensional grid of 45-degree segment (a) and grid from the middle of the photobioreactor (b).

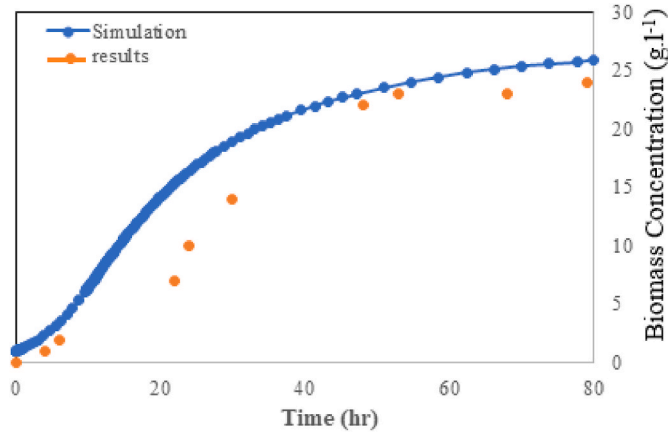


Fig. 4. The results of biomass concentration (g.l^{-1}) obtained from the simulation results and experimental data [25].

place in the bulk of fluid. The velocity field is represented by the vector u , which is derived by concurrently solving the momentum conservation equation in the mixture model and the mass continuity equation.

$$\nabla \cdot (-D_i \nabla c_i) + u \cdot \nabla c_i = R \quad (6)$$

Eq. (6) is also used to compute the mass transfer rate of gas components owing to the chemical reaction occurring in the fluid volume. The following is a list of other boundary and initial concentration conditions that may be found at the bed's inflow and outflow:

Since the wall of the photobioreactor is indiffusible, that is, Eq. (7).

$$n \cdot u_i = 0 \quad (7)$$

The relative composition of CO_2 and O_2 in the inlet gas is equal to 13.61 % and 16.1 %, respectively [25]. The boundary condition in the

inflow gas is of constant concentration type (Eq. (8)), according to the calculation of the inlet concentration at 37°C and 1 at m. (Table 3).

$$C_i = C_{i,0} = MD \times X_i \quad (8)$$

There is no mass diffusion of the fluid layer compared to its bottom layer along the bed axis (Eq. (9)):

$$-n \cdot D_i \nabla C_i = 0 \quad (9)$$

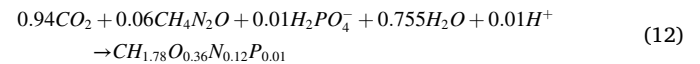
A unique constraint called the periodic condition is used to describe the liquid phase recirculation. As long as this boundary condition holds, the concentration of chemical constituents in the bed's liquid outlet flow (also known as the source flow) and the LFBPBR liquid inlet flow (also known as the destination flow) are equal (Eq. (10)).

$$C_{i,src} = C_{i,dst} \quad (10)$$

Besides, the following equation is established for both source and destination flows (Eq. (11)):

$$-n_{src} \cdot (J_i + u c_i)_{src} = n_{dst} \cdot (J_i + u c_i)_{dst} \quad (11)$$

The oxygen production efficiency, CO_2 consumption to produce the biomass and the stoichiometric coefficients are indirectly involved in determining the mass transfer rate of constituents based on the stoichiometric growth coefficients of *chlorella sorokiniana* microalgae with chemical composition ($\text{CH}_{1.78}\text{O}_{0.36}\text{N}_{0.12}\text{P}_{0.01}$) and the amount of nutrients in the culture medium (Eq. (12)) [28].



2.2.5. Radiation transport modeling

The modified Beer-Lambert model (Eq. (13)) was most suited for distributing and transmitting light from the light source to the photobioreactor [11].

Table 5

Investigation of grid independence test by comparing the numerical solution results of three different grids.

Row	The number of cells	Percentage increase	Microalgae biomass concentration (g.l^{-1})	Percentage increase of biomass concentration
1	108,173	–	21	–
2 *	121,435	12	25.8	23
3	686,409	465	26.4	2

$$I_{ph} = I_{ph0} \times \exp(-a_x \times C_{alg} \times \varphi_d \times Z), I_{ph0} = 1446 \times 10^{-6} \text{ mol.m}^{-2}.\text{s}^{-1} \quad (13)$$

φ_d and C_{alg} are used in Eq. (13) to highlight how the light flow is dependent on the model's other key physics. a_x is the light absorption coefficient and Z is the distance between each fluid point and the photobioreactor wall. The initial amount of light flux that was transferred to the substrate is denoted by the quantity I_{ph0} . Every single 3D point inside the bed at a constant and specified height has the coordinates of length (x) and width (y). Z is the difference in the bed radius (R) from this distance, and $(x^2 + y^2)^{0.5}$ is the distance that separates this supposed location from the origin of the coordinates (Eq. (14)).

$$Z = R - (x^2 + y^2)^{0.5} \quad (14)$$

2.3. Geometry of the system

The COMSOL Multiphysics software version 5.5a was utilized for the design of the geometric model in accordance to the schematic picture shown in Fig. 2 as well as the information that was readily accessible for the experimental design of LFBPBR. A three-dimensional schematic of the geometry with the appropriate dimensions (diameter of 10.4 cm and a height of 50 cm) is shown in Fig. 2. Both geometry and boundary conditions were symmetrical; which can be exploited to reduce the volume and cost of computations, especially in the formulation of boundary conditions, since it can be applied to both the flow field and mass transfer. Instead of analyzing the entire 360° cylindrical bed

(Fig. 2a), the computations can be performed on 45-degree segment (Fig. 2b).

2.4. Numerical methods and implementation

2.4.1. Grid characteristics

To numerically solve the mathematical equations governing the microalgae culture process inside the LFBPBR, the computational domain is discretized into small components called computational cells or meshes. Each equation is solved on these meshes, which are connected through boundaries. The software is capable of ideal execution of all stages before solving equations. In this research, meshing of the cylindrical bed of the LFBPBR is performed manually. Table 4 shows statistical analysis of the computing cells.

This segmentation features an intake flow, an output flow comprising both gas and liquid phases, a bed wall, and symmetrical boundary conditions on the lateral sides. A three-dimensional image of a 45-degree segment is shown and a portion of the meshing that occurs in the central zone of the LFBPBR is seen in Fig. 3. This meshing involves the wall (areas 1, 2, and 3) as well as the lateral sides (areas 4 and 5), and it also involves the volumetric meshes that exists inside the bed.

2.4.2. Model solution

After generation the geometry by COMSOL 5.5a software, numerical computations are performed according to the finite element method

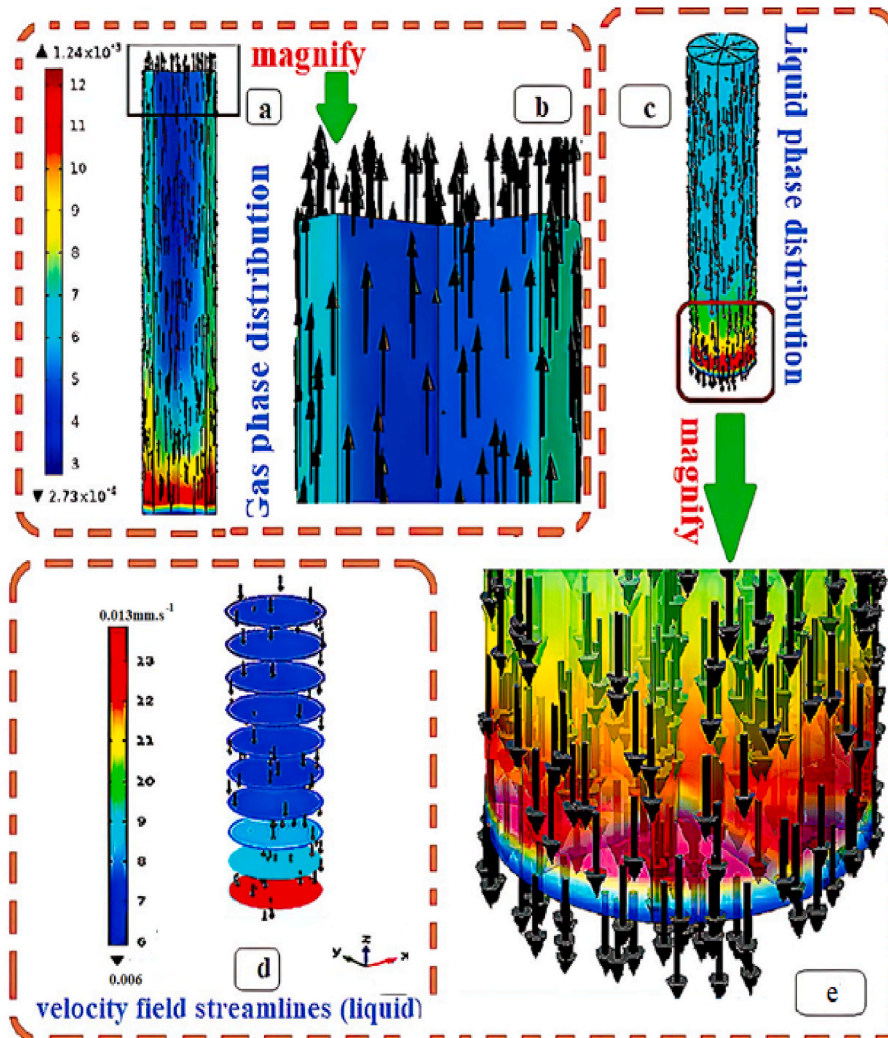


Fig. 5. Continuous (gas) phase (a,b) and dispersed (liquid) phase (c) and velocity distribution with streamlines (d and e) in mm.s^{-1} .

(FEM) and time step was choice equal 0.005 s. Computing and data processing was performed by a supercomputer, with 24 parallel processor cores and 96 GB of the read access memory (RAM), made by Super Micro Company under the Microsoft Windows Server.

3. Results and discussion

3.1. Confirmation of grid independence

The validity of the simulation results should be compared to the actual results. The experimental data used in this study were obtained from a scientific paper published in 2018 [25]. The simulation results of biomass concentration in g.l^{-1} are compared with the experimental results (by time) and are presented in the form of radiation transport (Fig. 4).

According to Fig. 4, the 10 experimental data are compared with numerical results. The relative error can be calculated from the following equation (Eq. (15)):

$$= \frac{|\text{experimental results} - \text{simulation results}|}{\text{experimental results}} \times 100 \% \text{ Percentage of relative error} \quad (15)$$

The relative error ranges from 12 to 26 % using the above equation. In Eq. (15), experimental and simulation values of algal biomass concentration are used and compared to calculate the relative error because biomass concentration data is fully available in the experimental reference [25] and is a good alternative for simulation validation. As can be seen in Fig. 4, the simulation values are always higher than the actual values, i.e., the simulation is an overestimate. The higher simulation values for biomass concentration compared to the experimental values can be attributed to the influence of the cell death rate (r_d) variable on the microalgae growth modeling. This means that the effect of this parameter on modeling has led to the production of overestimated data. Additional two-dimensional and three-dimensional curves are presented to provide further insights into the analysis of the microalgae growth dynamics. The validation of the results of the mass transfer model with real data has been investigated so far. Since the mass transfer model is obtained by solving the basic flow model, the Navier-Stokes (N-S) equations, if the accuracy of the simulation results of the mass transfer model is confirmed, the hydrodynamic model used will also be practically confirmed. This means that the equations and assumptions used to describe the two-phase fluid flow of the foam (the mixture model) provide reliable results. Numerical calculations were performed to evaluate the independence of the solution results from the number of computational cells using three different grids and then compared before performing the main experiment. This experiment shows the optimal number of meshes in a grid structure to achieve the desired simulation results, provided that the cost and computing power of the hardware remained unchanged. After finding the optimal number of cells, by fine-tuning the grid, the results of the solution practically do not change significantly, but the computational time and burden increase, which is undesirable (Table 5).

As can be seen in Table 3, the second row (denoted by *) is selected as the optimal and cost-effective number of cells and is the basis of numerical calculations throughout this simulation study.

3.2. Hydrodynamic profiles of the LFBPBR

The continuous and dispersed phases velocity fields, i.e., the air phase (in a foam component) along with the gas flow flux at different sections of the bed height can be seen in Fig. 5.

In Fig. 5a and b the direction of the velocity vector is upward, which corresponds to the physics of the problem. Considering the transfer of gases consumed by microalgae cells, i.e. O_2 and CO_2 , the pressure and velocity of the gas flow are high at the inflow. When gas molecules are

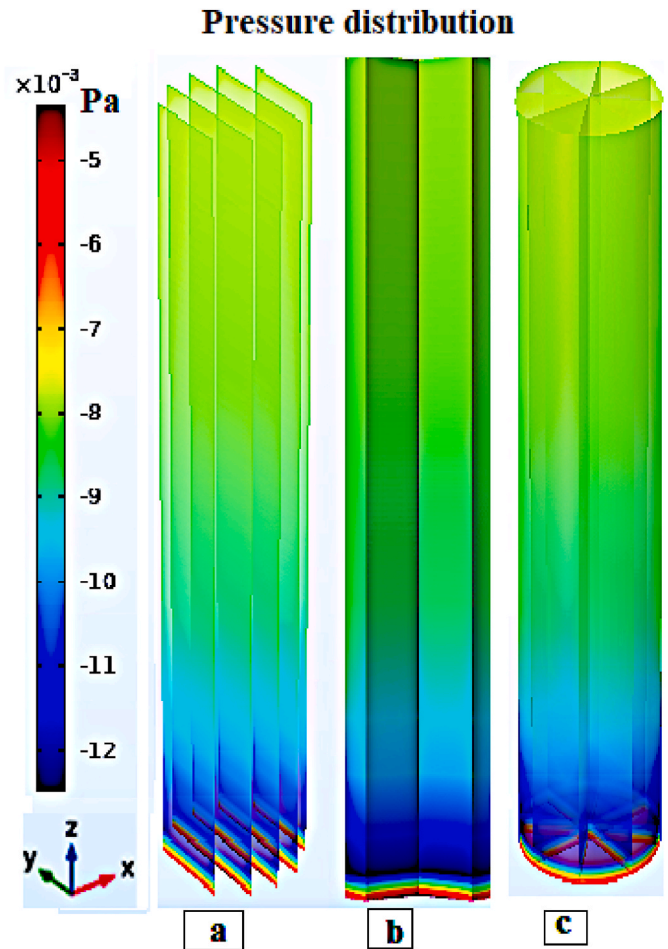


Fig. 6. Pressure distribution in the LFBPBR from parallel sectioned plates yz (a), inside view (b) and transparent view (c).

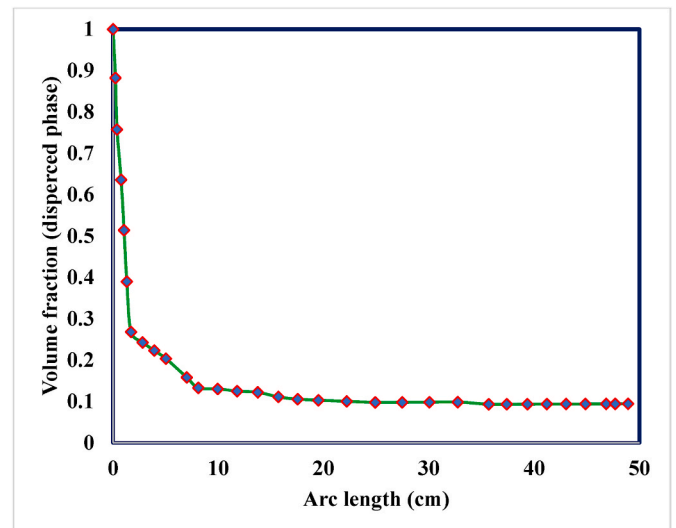


Fig. 7. Diagram of the volume fraction of the liquid phase at different heights of the LFBPBR.

trapped inside the foam gas phase and climb to the bed, their velocity gradually decreases in the axial direction of the column, with the maximum gas phase velocity at the bottom and the lowest velocity at the top of the column. Furthermore, Fig. 5c shows the liquid phase velocity

field. Based on the reflux flow in the column, the direction of fluid movement is downward. A close view of the liquid phase velocity field at the bed outlet is depicted in Fig. 5e; which elucidates the transparent view of the velocity field and the full view of the changes in the LFBPBR. According to Fig. 5d and e, the variations of the dispersed phase velocity are downward. Fig. 5d is shown with 50 vectors, for the liquid phase from top to bottom. The color of the sectional curves shows that the gas phase velocity increases from top to bottom and varies from about $0.0006 \text{ mm} \cdot \text{s}^{-1}$ to $0.0013 \text{ mm} \cdot \text{s}^{-1}$, respectively. Given the static solution of the Navier-Stokes equations, the velocity field and its corresponding vectors are expected to have no dependence on the time variable.

Fig. 6 illustrates the pressure field inside the photobioreactor (in Pa). Significant changes in the pressure field are observed primarily along the bed axis. Despite the changes in the variables are associated with

mass transfer and reaction, the radial changes are negligible here. The pressure varies from the bottom of the column containing the foam to the top. Gas flow is minimal at the injection site (4.33×10^{-3}). As the height increases, so does the hydrodynamic pressure. The hydrodynamic pressure in the air inlet is relatively higher and is equal to $0.01\text{--}0.012 \text{ Pa}$ due to the high pressure supplied by the pump to transfer the inlet gas flow to the bed. The hydrostatic pressure changes are minor due to the low density of the wet foam flow as expected based on to experimental results.

One of the important parameters is the volume fraction of the liquid phase. This indicates the amount of liquid is present in the space between the foams, which is important in determining and modeling the hydrodynamic behavior and mass transfer of the LFBPBR. As the bubbles move upwards along the bed due to contact with the reflux flow, the

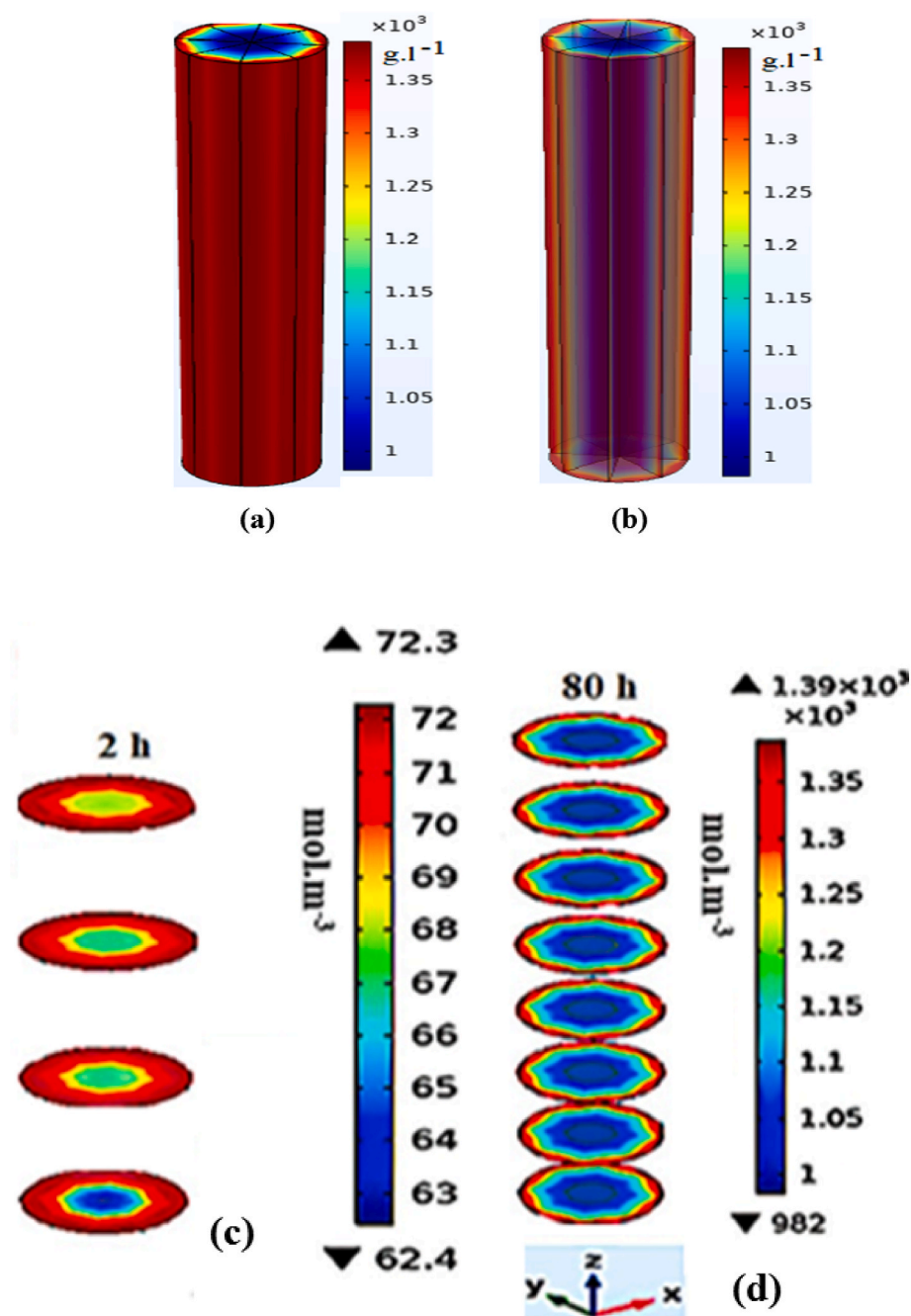


Fig. 8. Microalgae biomass concentration along with the height of the photobioreactor (a) 3-D, (b) 3-D with iso-surfaces in the radial direction (c) concentration distribution ($\text{mol} \cdot \text{m}^{-3}$) in surfaces in xy planes after 2 h (d) concentration distribution ($\text{mol} \cdot \text{m}^{-3}$) in surfaces in xy planes after 80 h.

stability of the foam decreases and causes the foam to decompose, and then the amount of liquid enclosed between the foams produced from the upper parts of the LFBPBR is more. The higher this hydrodynamic parameter means, the presence of more foam in that area and trapping more liquid in the space between the foams, and the possibility of more mass transfer between the liquid and gas phases is provided. In this condition, provided there is enough light, the photosynthesis reaction takes place and the efficiency of biomass production increases. At the bottom of the bed and at low heights (less than 10 cm), the liquid or microalgae culture medium is the highest and the volume fraction is equal to one. With the injection of air and the presence of light, the amount of trapped liquid reaches about 0.09 in the upper half of the bioreactor, resulting in a decrease of this variable with increasing height. One reasons for the reduction of liquid trapped between the foams is the less tendency to form stable foams at higher bed heights. Fig. 7 illustrates the graph of this variable, at low heights there is the largest amount of liquid between the foams and up to a height of 3 cm, nearly 71 % drop and it reaches about 29 % and at a height of 10 cm, the reduction of this variable reaches 87 %. Identification of volume fraction of liquids leads to provide deep insight in the photobioreactor efficiency. Since major mass transfer relates to the gas liquid contact area followed by high volume fraction of liquids. Thus, one can find optimum reactor design condition by deep analysis of volume fraction results. For example, in this research, major volume fraction reduction (87 %) occurs at the 10 cm of the photobioreactor which can be great choice for photobioreactor design. Beside this understanding, economical factors play a vital role for reactor design which should be consider in future researches.

3.3. Concentration profiles in the LFBPBR

The three-dimensional profile for biomass concentration in the photobioreactor after solving the equations and can be display in Fig. 8a, b, 8c, and 8-d.

The intensity of received light flux and the microalgae growth are maximum close to the surface or wall. Closer to the center of the bed, low light intensity alters chloroplast ultrastructure and photosynthetic

metabolism; meaning that the inner layers of the microalgae biomass accumulate on top of each other and receive less light. Microalgae concentration at 8 different heights of the bed 80 and 2 h after the experiment are detailed in Fig. 8. According to Fig. 8c the maximum concentration of microalgae (72.3 mol m^{-3}) is produced close to the wall after 2 h, and this amount reaches 1390 mol m^{-3} (Fig. 8d) after 80 h.

During the initial stages of the process, before cells entering the logarithmic growth phase, the growth rate values are positive. Subsequently, the living cells in the culture medium are constantly produced, and the production rate is positive. These values are logical compared to the delayed phase, because in the logarithmic growth phase more cells multiply and grow per unit of time. But in some areas close to the center, the production rate of microalgae cells is negative, indicating that some cells not only do not grow, but also a small number suffer cell death. Lower intensities of light are accessible in the inner areas of the bed; leading to an increased possibility of cell death and growth inhibition. Initial stages of the growth are observed to provide lower growth rates while the range of changes in the cell death rate is relatively low. With the increase in the cell growth rate due to better access to the carbon source and light energy, the number of dead cells increases per unit of time.

In Fig. 9a and b whose results shows the changes in the microalgal biomass concentration. The cell density produced, 22 g l^{-1} , is minimal in the center, and the maximum biomass produced is 30.9 g l^{-1} at the bed wall after 80 h (Fig. 9b).

Regarding the radial direction of the concentration of CO_2 , due to higher availability and accessibility of light on the surface of the LFBPBR, the consumption of the carbon source and the cell efficiency is higher in the parts close to the wall. As it gets closer to the center of the cylindrical bed, the concentration of carbon dioxide gas increases or is consumed at a lower rate. Besides, there are few changes in the axial direction of CO_2 gas changes, because the light distribution along the bed axis is small. It seems that the factor of biological reaction and access to light is important in determining the radial changes of the gas, and this importance is not seen in the axial direction.

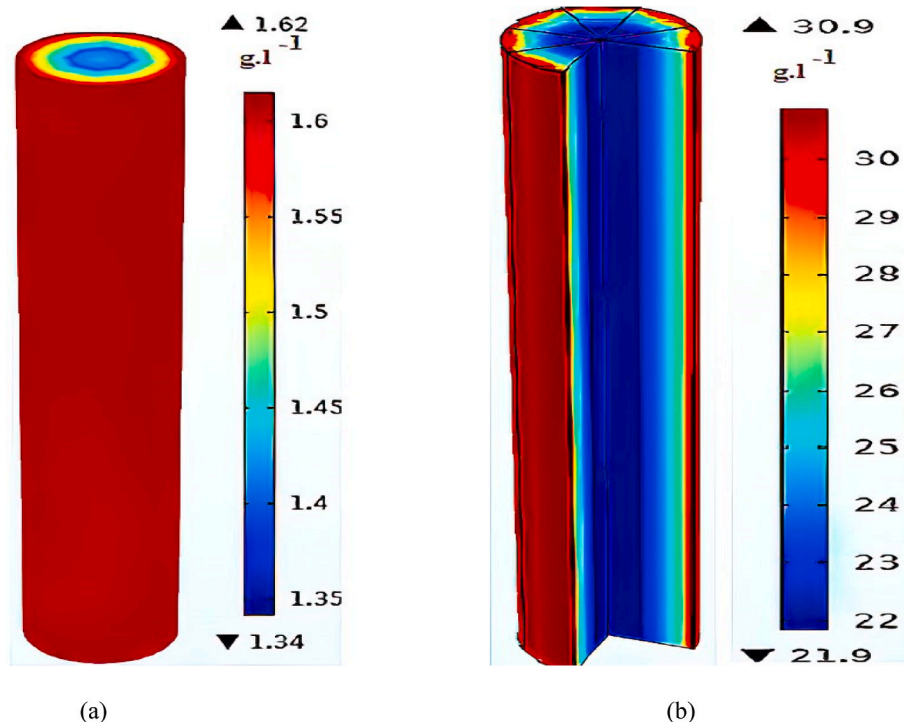


Fig. 9. Microalgal concentration distribution (g.l^{-1}) after (a) 2 h, (b) 80 h.

4. Conclusion

In this work, a three-dimensional model was developed to numerically investigate the bioprocess of microalgae production using CFD technique. The simulation results were satisfactory, compared to actual experimental data; with relative errors below 26 %. Therefore, the assumptions made are seemingly proper and the suggested model is correct for hydrodynamics and mass transport. Despite the results obtained in this study for concentration of microalgae biomass (up to 30.9 g l^{-1} microalgae cell mass production), the issue of optimizing the light distribution in different layers of cylindrical photobioreactors is a challenging. Because only a limited sections of the available areas of the LFBPBR (distances of (from) 3–5 cm inside the bed) will be used to produce cells. The limiting factor in microalgal biomass producing is the light availability both on the surface and in the internal sections. The radiation flux received by the cells decreases non-linearly at different times from the start of the bioprocess. The received light flux and the growth of microalgae are maximum on the surface of the bed, and the more central areas are effectively low on microalgae concentration due to the inadequate light supply.

Future studies are recommended to achieve more accurate solutions. The Euler-Euler (EE) model can be used instead of the mixture model and the results can be compared with the experimental results. Also, the Lagrangian approach can be used to study the path of a single bubble or population of bubbles within the liquid foam-bed photobioreactor. Besides, the bubble population balance equation (PBE) added to the set of mathematical equations can be solved in the computational domain. The particle tracking module can also be proper for this approach. Based on accuracy and calculation demand of simulation of flow of two phase that

consist of gas bubble and liquid flow Lagrangian model simulate each bubble flow in liquid and are have the best accuracy with high calculation demand, EE model consider the motion of bubble like continuous phase with independent velocity that can accounts the acceleration and deceleration of bubble in fluid flow in contrast in mixture model constant velocity difference (slip velocity) between liquid and gas are considered and no independent velocity profile or bubble are calculated.

CRediT authorship contribution statement

Seyyed Javad Heydari: Performing simulations, Investigation, Data curation, Formal analysis, Methodology, Writing original draft. **Arezou Jafari:** Supervision, Formal analysis, Methodology, Funding acquisition, Review and editing original draft. **Seyyed Mohammad Mousavi:** Supervision, Validation, Conceptualization, Project administration, Funding acquisition, Review and editing original and revised version. **Mahdi Abdi Khanghah:** Validation, Review and editing revised version.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This study was financially supported by Tarbiat Modares University under grant number IG-39701.

Nomenclature

Symbols and abbreviations

T	bed temperature ($^{\circ}\text{C}$)
$u_{g,\text{in}}$	gas input velocity (m.s^{-1})
$u_{l,\text{in}}$	liquid input velocity (m.s^{-1})
R	the LFBPBR radius (m)
X_{CO_2}	input CO_2 percentage (%)
X_{O_2}	input O_2 percentage (%)
I_{ph}	light flux ($\mu\text{mol.m}^{-2}.\text{s}^{-1}$)
$I_{\text{ph},0}$	initial light flux ($\mu\text{mol.m}^{-2}.\text{s}^{-1}$)
MD	molecular mass of air (g.mol^{-1})
a_x	light specific absorption coefficient ($\text{m}^2.\text{mol}^{-1}$)
D_{O_2}	oxygen diffusion coefficient ($\text{m}^2.\text{s}^{-1}$)
D_{CO_2}	carbon dioxide diffusion coefficient ($\text{m}^2.\text{s}^{-1}$)
P_t	total pressure (Pa)
P	hydrodynamic pressure (Pa)
C_i	inlet concentration of gas constituents (mol.m^{-3})
j_{slip}	relative velocity vector (mm.s^{-1})
$u_{\text{slip},x}$	interphase velocity (mm.s^{-1})
F_m	interphase momentum transfer Force (N)
F	external force (N)
m_{dc}	interphase transfer rate (kg. s^{-1})
N_2	Nitrogen
R	Reaction term ($\text{mmol.m}^{-3}.\text{s}^{-1}$)
VOF	Volume of fluid (m^3)
FEM	Finite element method
EE	Euler-Euler
RAM	Random-access memory
N–S	Navier–Stokes
atm	atmosphere
n	normal vector
LFBPBR	Liquid Foam Bed-photobioreactor
CFD	Computational Fluid Dynamics

Greek symbols

μ_l	liquid viscosity (s^{-1})
ρ_d	dispersed phase density ($kg.m^{-3}$)
ρ_c	continuous phase density ($kg.m^{-3}$)
φ_d	dispersed phase volume fraction

Subscripts

d	dispersed
c	continuous
l	liquid
O ₂	oxygen
CO ₂	carbon dioxide
ph	photon
srs	source
des	destination

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